

Wi-Fi Calling System (LAN-Based Voice Communicator)

A decentralized, real-time voice communication application that enables high-quality audio calls between devices on the same local network — no internet required.

About The Project

Most voice calling apps depend on external servers and internet connectivity. This project solves that problem by building a fully offline, LAN-based calling system using WebRTC for audio communication and custom TCP/UDP signaling for session management.

This application was built as a demonstration of low-level networking, real-time communication protocols, and mobile application development.

Features

- Real-time voice calls over a local Wi-Fi network
- Automatic peer discovery using UDP broadcast — no manual IP entry needed

- WebRTC-based audio with low latency and high quality
- TCP-based signaling for reliable SDP and ICE candidate exchange
- Fully decentralized – no server, no internet, no third-party dependency
- Secure offline communication architecture

Tech Stack

Layer	Technology
Mobile App	Flutter, Dart
Audio Communication	WebRTC
Peer Discovery	UDP Broadcast
Session Signaling	TCP Sockets (SDP, ICE)
Architecture	Decentralized P2P

How It Works

1. App starts and broadcasts a UDP discovery packet on the local network
2. Other devices on the same Wi-Fi detect the broadcast and respond

3. A TCP connection is established between peers for signaling
4. SDP (Session Description Protocol) and ICE candidates are exchanged over TCP
5. WebRTC takes over and establishes a direct peer-to-peer audio stream
6. Voice call begins — fully offline, no server involved

Project Structure

```
wifi_calling_system/  
├── lib/  
│   ├── main.dart          # App entry  
point  
│   ├── discovery/        # UDP  
broadcast and peer detection  
│   ├── signaling/        # TCP-based  
SDP and ICE exchange  
│   ├── webrtc/           # WebRTC audio  
session management  
│   └── ui/               # Call screen  
and peer list UI  
├── pubspec.yaml          # Flutter  
dependencies  
└── README.md
```

Setup & Run

Prerequisites

- Flutter SDK installed
- Two devices connected to the same Wi-Fi network
- Android device or emulator (API 21+)

Steps

```
# Clone the repository
git clone
https://github.com/atulpandey2626/wifi-
calling-system.git

# Navigate to project folder
cd wifi-calling-system

# Install dependencies
flutter pub get

# Run on connected device
flutter run
```

Run the app on two devices connected to the same Wi-Fi. Device A will discover Device B automatically and initiate a call.

Key Learnings

- Practical implementation of WebRTC in a mobile environment
- Low-level UDP and TCP socket programming in Dart

- Peer-to-peer architecture design without central servers
- Real-time audio streaming and latency optimization

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